

Homework # 3

PHY-905-003

Computational Astrophysics and Astrostatistics

Spring 2026

This assignment is due by 11:59 p.m. on Friday April 10, 2026.

Instructions: Turn in all materials via GitHub. Include your code, plots, and answers to any questions asked in your assignment. Your code must adhere to the class coding standards and use text files rather than Jupyter notebooks. The solutions to individual parts of the assignment should be saved in separate, clearly-named subdirectories named `part_1`, `part_2`, etc. Plots should have easily readable and logical axis labels and titles, and the source code and data used to generate the plots should be included. Questions should be answered in a file in each subdirectory named `ANSWERS.md` or in a $\text{\LaTeX}$ -created PDF document of a similar name (e.g., `ANSWERS.pdf`).

Part 1: You have been asked to do a spectral analysis of a time series observation from an X-ray timing instrument, which can be found in the file `part_1_files/instrument_data_output.dat`. This instrument is having some technical issues and periodically has to shut down and reboot itself in order to continue working¹, so some of the data is missing and the signal has been set to zero. To help you in identifying the parts of the observation where the errors occur the instrument engineers have helpfully set an error code in the file². Inspection of the signal shows that it is likely to be comprised of a small number of time-varying components. **The goal of your analysis is to identify and report the frequencies, amplitudes, and phases of the signal components and create a synthetic (noise-free) signal that you can compare to the dataset.** In addition to reporting that information (in `ANSWERS.md` or another similar file), make sure to include a comparison between your synthetic signal and the observation and a discussion of the likely source of differences between the two. Include any plots that are necessary to demonstrate your results. Also, describe in your writeup if there any features in the dataset that are missing from your synthetic signal. If you had to adjust your synthetic signal by hand in some way to make them agree better, what did you do?

Here are some suggestions/questions that might be useful when constructing your analysis and your report:

- Remember that `scipy.fftpack.fft` returns complex numbers that encode both amplitude and phase information – in other words, recall from your pre-class readings that the “real” component of the signal is in phase with (has a $0^\circ/0$ radian phase shift from) a reference signal, and the “imaginary” component is $90^\circ/0.5\pi$ radians out of phase with the reference signal. An arbitrary signal at a given frequency can have both real and imaginary components, which indicates a phase shift that is neither 0° or 90° from the reference signal. Refer to [this useful Stack Overflow post](#) for one way of extracting the amplitude and phase information from the array, and think about how that would impact the synthetic signal that you’re constructing.
- If the frequencies where you expect the oscillations are relatively high compared to the temporal length of the observation (i.e., $1/f_{osc} \ll T_{obs}$), do you need to analyze the entire observation to extract that information?
- Are there types of time-varying behaviors that would not show up in a Fourier analysis? If so, what are they?

¹The engineers blame the errors on “cosmic rays,” but that’s what they blame everything on so you’re somewhat skeptical.

²The comments at the top of the file describe the columns of data.

Part 2: The [Traveling Salesman Problem](#) is a classic and challenging problem in computer science where the focus is on optimization. The problem is as follows: Imagine there is a salesman who has to travel to N_C cities. The order is unimportant, as long as they only visit each city once on each trip, and finishes where they started. The salesman wants to keep the distance traveled (and thus travel costs) as low as possible. This problem is interesting for a variety of reasons - it applies to transportation (finding the most efficient bus routes), logistics (finding the best UPS or FedEx delivery routes for some number of packages), or in optimizing manufacturing processes to reduce cost. In astrophysics, it is akin to model parameter optimization. The Traveling Salesman Problem is extremely difficult to solve exactly for large numbers of cities - testing every possible combination of cities would take $N_C!$ individual tests. As a result, methods to quickly find a “good enough” solution are of great interest.

In this problem, you are going to use the code in `part_2_files/traveling_salesman.py` to minimize the traveling salesman problem for $N_C = 30$ cities to find a “good enough” solution. Your code should take some number of iterations, doing the following at each step:

1. Randomly swap N_S contiguous cities in the array of cities - in other words, if you have cities (1,2,3,4,5,6,7,8,9,10) for $N_S = 2$ you could swap two pairs of cities to get (1,2,8,9,5,6,7,3,4,10). N_S is typically 1, but can be more than 1 if you wish.
2. Check the total distance traversed by the salesman.
3. If the new ordering results in a shorter path, keep it. If not, throw it away.

You stop when you attain convergence. Decide what that means in this context and make sure to report it! Also keep track of the steps and the minimum distance traveled as a function of number of iteration and plot out the minimum distance as a function of step. (If you’re feeling ambitious, make a movie of what it looks like as your simulation evolves toward the minimum path.)

Some questions to answer/tests to do:

1. How do you decide when to stop?
2. Does varying N_S from 1-5, but keeping it constant per run, significantly affect the rate of convergence?
3. Does allowing N_S to randomly vary from 1-5 every time step in a single run significantly speed convergence?
4. Assuming you can test one model per microsecond per core using ICER’s High Performance Computing Center (which has approximately 7×10^4 CPU cores across several clusters), how long would it take to test every single combination for $N_C = 30$ to get the provably best solution, and how does that compare to how long it takes to get a reasonably good solution using your implementation?

Note that you should do many trials per choice of N_S to get a sense of the typical behavior! Regarding trials, it’s very important to recall that Python’s [random](#) library uses the [Mersenne Twister](#) algorithm, which is a [pseudorandom number generator](#) – it uses an algorithm that generates a sequence of numbers that are *statistically random*, but which are repeatable given the same initial number – in other words, the sequence of numbers has properties that approximate a truly random sequence of numbers, but the order of numbers is *deterministic*. This means that you need to change the initial “random seed” number to get a different sequence of ensuing numbers!

Part 3: In this section you’re going to speed up the Python code that you wrote in Part 2 of this homework problem using the parallelism techniques you learned in class. To that end, do the following (and document timing and other progress along the way with tables, figures, and/or text):

1. Establish a performance baseline by timing the single-core version of your code for a variety of different numbers of cities (from just a few cities up to 100 cities). Also use timers (from the Python [time library](#) – e.g. `time.perf_counter()`) to identify the costly part(s) of your code, as an opportunity for future optimization. How does the cost change? Hint: run multiple trials and change the random seed in the random number generator to get a sense of performance variability as a function of system load (i.e., what other processes are running on the CPU core) and the number of iterations at a given cost.
2. Examine your code and look for serial “hot spots” – i.e., places where the code is taking significant fractions of the computing time in its current serial (single CPU core) form. It’s useful to do this for different numbers of cities – as you change that, you’ll see that due to the algorithmic complexity of the problem different parts will change in cost while others remain the same!
3. Try to speed up the hot spots by making simple fixes – for example, try to find places where you can replace lists with NumPy arrays, loops with NumPy operations, etc. Test those fixes with additional timing tests and document what you did and how it changed the code’s speed. Note that if you did a good job writing your code you may not see much improvement here!
4. Try accelerating your serial code using [Numba](#) decorators. How much does this accelerate your code for different numbers of cities for a given random seed (i.e., a given number of iterations per number of cities)? If you try Numba’s [parallelization](#) options, what happens? (Hint: think about what Numba does to parallelize this problem – how easily can it be parallelized as you’ve currently formulated it?)
5. Now accelerate the code by using the Python [multiprocessing](#) library (do this separately from your Numba effort). Focus your parallelism efforts on the hot spots you identified. Document the speedup compared to the serial case. Hint: how would you parallelize this optimization problem? Which parts of the problem do you need to distribute to different CPU cores, and how would you do that? Also, what do you need to do with regard to the random seed(s) you use for optimization? (Think about what would happen if every CPU core uses the same random seed - what would you expect to happen?)

Submit the various versions of the code that you’ve written in clearly-named files. Also, make sure to write a narrative explaining the things that you’ve done to speed up your code and the results that you see. For the problem you’re trying to accelerate, is there an optimal number of CPU cores where you get the best speedup? How does that vary as a function of the problem size and optimization technique?

Hint: You’re going to need Python *partial functions* for the Multiprocessing component of this option. Check out [this tutorial](#) on partial functions. Here’s an example of how to use a partial function:

```
my_partial = functools.partial(my_func, arg1=val1, arg2=val2)

with multiprocessing.Pool(processes=nprocs) as p:
    results = p.map(my_partial, work_to_share)
```

Alternately, you can use `mutliprocessing.Pool().starmap`, which is similar to the function taught in class (`multiprocessing.Pool().map`) but handles functions with multiple arguments. This option is useful if you don’t want every process to run your desired function with the same arguments (as will happen when using a partial function).

Part 4: “Stellar population synthesis” models are a broad class of astrophysical models that are used to make predictions about (and understand observations probing) phenomena relating to stellar populations. This can include understanding the metallicities, ages, star formation histories, and initial mass functions of stars in galaxies ([Evolutionary population synthesis](#)³) or predicting the behavior of the population of compact objects that are the end products of these stellar populations, such as their emission of gravitational waves as a function of time due to binary black hole mergers ([Binary population synthesis](#)⁴). **Your goal is to make a simple binary population synthesis code that will predict the behavior of a population of stars over time** – in particular, the rate of Type Ia and Type II supernovae and mergers of neutron stars and black holes (and the resulting gravitational wave signatures) as a function of time.

To do this, start from the stellar initial mass function sampling code you used in class, which creates a list of stars comprising a given stellar cluster given a set of IMF parameters. After you generate that code, modify it to calculate the post-main-sequence behavior of the population as a function of time. Do that using the following assumptions⁵:

- Stars have a mass-dependent main sequence lifetime of $t(M_*) = 10(M_*/M_\odot)^{-2.5}$ Gyr, with a minimum lifetime of 2 Myr (at the highest-mass end, the most massive stars take roughly a constant amount of time to deplete their cores of fuel).
- Stars have a mass-dependent binary fraction (i.e., probability of being found in a gravitationally-bound pair). 75% of stars of $\geq 8 M_\odot$ are found in binaries; 50% of stars with lower masses are in binaries.
- Stars are most likely to form in binary systems with companions of similar masses – to that end, assign all stars in binaries to a companion whose mass is within 50% of its mass, if possible. If that’s not possible, assign it to the star with the closest mass you can.
- All stars with $m_* \geq 8 M_\odot$ explode as Type II supernovae, which you should assume happens promptly at the end of their main sequence lifetimes; of these supernovae, 80% of these supernovae will result in neutron stars and 20% will create black holes as compact remnants.
- All stars below $8 M_\odot$ evolve into asymptotic giant branch (AGB) stars at the end of their main sequence lifetimes, and ultimately shed their stellar envelopes and become white dwarfs. Assume this happens promptly at the end of their main sequence lifetime.
- Any binary system that has a neutron star-neutron star, black hole-black hole, or neutron star-black hole pair is assumed to merge due to gravitational radiation. This takes time, because the orbital energy needs to be removed by gravitational radiation. Assume that the delay time for any individual pair of stars to merge is a lognormal distribution with a peak of 1 Gyr and a width of 200 Myr, with the time delay occurring after the end of the main sequence lifetime of the less massive star in the pair.
- Any binary system that has a white dwarf-white dwarf pair is assumed to explode in a Type Ia supernova promptly at the end of the main sequence lifetime of the less massive star in the pair.
- Any star that is not in a binary system, or that does not conform to one of the categories described above, is assumed to end its main sequence lifetime in one of the ways described above but do nothing else that is interesting for the purpose of this discussion.

³As exemplified by [FSPS](#); see Conroy’s papers – [Paper 1](#), [Paper 2](#), [Paper 3](#)

⁴As exemplified by the [COSMIC](#) code; see [Breivik et al. 2021](#)

⁵Note that these assumptions are significantly simpler than in standard binary population synthesis codes, but capture the spirit of what those codes can do. If you’re interested, talk to your instructor about the ways that a cutting edge research-grade binary population synthesis code deviates from the model you’re creating here.

Given these assumptions, assume a star cluster with a total mass of $10^3 M_{\odot}$, create a list of individual stars, and probabilistically assign them to binary systems. Calculate the instances of Type Ia and Type II supernovae and compact object mergers as a function of time (keeping track of the NS-NS, BH-BH, and NS-BH categories separately). Do this for a large number of clusters. Answer the following questions, including plots as necessary to make your points:

1. When do you expect to see Type Ia vs. Type II supernovae as a function of time? Does either of these quantities vary significantly between individual star clusters, and why?
2. When do you expect to see gravitational wave signatures coming from mergers of massive compact object pairs? (“Massive” in this context means comprised of neutron stars and/or black holes.)
3. What is the typical number of massive compact object pairs in a single star cluster, and how does this vary between clusters?
4. What fraction of the stellar population as a function of number of stars and, separately, of stellar mass explodes in a supernova and/or has a binary merger of some sort? In other words, what fraction of the stars do something “interesting”?
5. As you go to larger stellar populations (say, to star clusters of total mass 10^4 or $10^5 M_{\odot}$), does the cluster-to-cluster variation increase, decrease, or stay the same? Why do you think that is?

Part 5: It has been hypothesized that the stellar initial mass function may have an environmental dependence. To test this, you are being provided with two data files (in the directory **part_5_files**) containing a list of measured stellar masses for two separate star clusters – one toward the galactic anticenter (**IMF_1.dat**) and the other very close to the center of the galaxy (**IMF_2.dat**). Given the distances involved, only the masses of stars with masses greater than $2 M_{\odot}$ could be measured. Using the statistical techniques that you’ve learned this semester, (1) argue whether or not the stellar mass distributions of the two star clusters appear to have the same mass function, and (2) quantify your confidence in this result. Clearly explain the statistical techniques that you have used for both of these things! **Complicating factors:** Note that the star clusters have substantially different total masses and somewhat different (non-zero) ages. How might this affect your approach and/or your interpretation of the results?